

BSCIT 375: Knowledge Discovery and Data Science Assignment-II

Submit a handwritten solution for each of the following questions.

Show all steps in calculations where applicable.

- Q1 Distinguish between Knowledge Discovery in Databases (KDD) and Data Mining. Provide a real-world example of each.
- Q2 List and briefly describe the six phases of the CRISP-DM methodology.
- Q3 Define the following data types with examples: nominal, ordinal, continuous, discrete.
- Q4 Calculate the mean, median, and mode for the dataset: $\{12, 15, 12, 18, 22, 15, 19, 15\}$.
- Q5 Compute the range, variance, and standard deviation for the sample: $\{5, 10, 15, 20, 25\}$.
- Q6 A dataset has mean = 100 and standard deviation = 15. Find the Z-score for a value of 130. Is it an outlier under $|Z| > 3$ criterion?
- Q7 Explain the difference between Min-Max normalization and Z-score standardization. When would you prefer one over the other?
- Q8 Apply Min-Max normalization to the values $[50, 100, 150, 200]$ to scale them to $[0, 1]$.
- Q9 Define MCAR, MAR, and MNAR with examples.
- Q10 Using the IQR method, find outliers in: $\{10, 12, 14, 18, 22, 24, 30, 35, 100\}$. Show Q1, Q3, IQR, and fences.
- Q11 Calculate the Pearson correlation coefficient for $X = [1, 2, 3, 4, 5]$ and $Y = [2, 4, 6, 8, 10]$. Interpret the result.
- Q12 Given the contingency table below, compute the chi-square statistic and test independence at $\alpha = 0.05$ (critical value 3.841).
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|--------|------|------|
| | Pass | Fail |
| Male | 30 | 20 |
| Female | 25 | 25 |
- Q13 Compare one-hot encoding and label encoding. When is each appropriate?
- Q14 Explain Principal Component Analysis (PCA) in simple terms. How does it reduce dimensionality?
- Q15 Apply equal-width binning with 3 bins to the data: $[5, 7, 12, 15, 18, 22, 25, 30, 35, 40]$.
- Q16 A fair die is rolled. What is the probability of getting an even number or a number greater than 4?

- Q17 A committee of 4 is to be chosen from 8 men and 6 women. How many ways if it must have at least 2 women?
- Q18 Two cards are drawn without replacement from a standard deck. Find the probability both are hearts.
- Q19 Define conditional probability. If $P(A) = 0.6$, $P(B) = 0.4$, $P(A \cap B) = 0.2$, find $P(A|B)$ and $P(B|A)$.
- Q20 A test for a disease is 99
- Q21 A sample of 50 students has a mean test score of 75 with $\sigma = 10$. Construct a 95
- Q22 A sample of 20 items has mean 50 and sample standard deviation 5. Construct a 90
- Q23 In a survey of 200 people, 120 support a new policy. Find a 95
- Q24 A company claims that less than 10
- Q25 Given $X = [1, 2, 3, 4, 5]$ and $Y = [2, 4, 5, 4, 5]$, calculate the regression line $\hat{Y} = a + bX$.
- Q26 Compute the coefficient of determination (R^2) for the model in Q25.
- Q27 For the multiple regression model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$, explain the meaning of each coefficient.
- Q28 A house price model: $\text{Price} = 50 + 0.2 \times \text{Sqft} + 10 \times \text{Bedrooms}$. Predict price for 2000 sqft and 3 bedrooms.
- Q29 Compare supervised and unsupervised learning with one example of each.
- Q30 What is the K-Nearest Neighbors algorithm? Explain the role of K and distance metric.
- Q31 Given points $A(1, 2)$, $B(2, 3)$, $C(3, 3)$, $D(6, 7)$ and new point $P(2, 2)$. Using $K = 3$ and Euclidean distance, classify P if classes are $\{A, B, C\}$ as 'X' and D as 'Y'.
- Q32 State the Naive Bayes assumption. Why is it called "naive"?
- Q33 A dataset has 9 Yes and 5 No. Compute the entropy.
- Q34 Calculate Gini impurity for a node with 4 Yes and 6 No.
- Q35 Explain bagging. How does Random Forest improve upon a single decision tree?
- Q36 Describe the K-Means clustering algorithm step by step.
- Q37 Given points: $(1, 1)$, $(1, 2)$, $(2, 1)$, $(5, 5)$, $(5, 6)$, $(6, 5)$. Run one iteration of K-Means with $K = 2$ and initial centroids $(1, 1)$ and $(5, 5)$. Show assignments and new centroids.
- Q38 Define support, confidence, and lift for an association rule $X \Rightarrow Y$.

- Q39 Given $\text{support}(\{Milk\}) = 0.6$, $\text{support}(\{Bread\}) = 0.5$, $\text{support}(\{Milk, Bread\}) = 0.3$. Compute confidence and lift for $\{Milk\} \Rightarrow \{Bread\}$. Interpret lift.
- Q40 From the transaction data: T1: $\{A, B, C\}$; T2: $\{A, B\}$; T3: $\{A, C\}$; T4: $\{B, C\}$; T5: $\{A, B, C\}$, find all frequent itemsets with $\text{min_sup} = 0.6$.
- Q41 From the confusion matrix: TP=80, FN=20, FP=10, TN=90. Compute accuracy, precision, recall, and F1-score.
- Q42 Why is accuracy not a good metric for imbalanced data? Give an example.
- Q43 What is the ROC curve? Define TPR and FPR.
- Q44 What is the bias-variance trade-off? Explain with a sketch.
- Q45 Describe cross-validation. Why is stratified k-fold preferred for classification?
- Q46 Compute the dot product of $\mathbf{u} = [1, 2, 3]$ and $\mathbf{v} = [4, 5, 6]$.
- Q47 Find the Euclidean norm of vector $\mathbf{w} = [3, -4, 12]$.
- Q48 Compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f(x, y) = 3x^2y + \sin(xy)$.
- Q49 Write the chain rule for a composite function $h(x) = f(g(x))$. Compute the derivative of the sigmoid function $\sigma(z) = 1/(1 + e^{-z})$.
- Q50 Explain gradient descent update rule: $w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}$.

End of Assignment